

Chia Hui Yen

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PROFESSIONAL SUMMARY

Software engineer building backend and AI-powered systems for complex real-world workflows. Experienced in API services, retrieval pipelines, schema-validated extraction, and deployable ML-powered applications, with a focus on latency, reliability, and production-oriented system design. Turns ambiguous, unstructured inputs into robust software services and decision-support tools.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Computational Design (Focus: Applied / Production AI Systems)

May 2026

- **Relevant Coursework:** ML in Production, Principles of Software Construction, Data Structures & Algorithms, Intro to Deep Learning, WebApp Development. **Award:** CMU Architecture Merit Scholarship.

Tsinghua University

Beijing, China

Bachelor of Architecture

June 2024

- **Award:** Tsinghua University Malaysian Outstanding Undergraduate Scholarship.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, HTML/CSS.

Backend / Data Systems: FastAPI, AsyncIO, Pydantic, JSON Schema, ETL, PostgreSQL, Neo4j, Kafka.

Infrastructure / Deployment: Docker, AWS (EC2, S3, RDS), GitHub Actions, Kubernetes, Prometheus, Grafana.

AI / Retrieval Systems: PyTorch, Hugging Face, Embeddings, RAG, FAISS, BM25, Hybrid Retrieval, Reranking, LangChain, LangGraph, LLM APIs, Evaluation Harnesses, LoRA/PEFT.

WORK EXPERIENCE

Lennar Corporation

San Francisco, CA

Software Engineer Intern (AI/ML)

June 2025 – August 2025

- Built **FastAPI** microservices for image+text construction ingestion using **AsyncIO** concurrency, async batching, semaphore rate limiting, and retry/timeout controls for external LLM APIs; processed **1,000+ images in 240s** with stable throughput for high-volume field content workflows.
- **Improved retrieval over construction artifacts**, implemented a **LangChain + FAISS** retrieval layer with hybrid ranking (**keyword + embeddings + reranking**) and metadata indexing, enabling **sub-second** semantic search across construction documents and field-generated content, lowering search friction and reducing manual QC triage.
- **Structured noisy field communication into QC reports**, productionized an NLP pipeline that converted unstructured site conversations into schema-validated issue reports using few-shot classification, clustering, and **RAG** enrichment; generated **20,000+ embeddings in 300s** and reduced downstream data handling overhead through typed outputs.
- Raised extraction reliability and domain accuracy, improved information quality through **construction-aware chunking**, **keyword expansion**, curated few-shot prompting, and an **F1-tracked** evaluation harness; enforced typed outputs with **Pydantic** validation for safer downstream consumption.

PROJECTS

AEC Interpreter System | *Multimodal Fine-Tuning, LangGraph* | [GitHub Repo](#)

February 2026 – April 2026

- Engineered an IFC-based synthetic data pipeline with **IfcOpenShell**, **Blender**, and **Gemini** to generate **1,377** multimodal cases from BIM geometry, floorplans, rendered imagery, and fragmented field chats under low-data AEC constraints.
- Adapted **Qwen2.5-VL-7B** with **LoRA** for efficient fine-tuning to map noisy site inputs into typed spatial constraints, improving **Top-1 accuracy from 25.7% to 35.3%** and valid search-space retention from **52.8% to 66.2%** for BIM element localization from multimodal field evidence.
- Designed a hallucination-resistant neuro-symbolic retrieval pipeline that compiled schema-constrained outputs into **Cypher** over a **Neo4j** IFC graph, separating perception, query planning, and execution into modular components for auditable, topology-grounded BIM retrieval.
- Benchmarked a **LangGraph ReAct + FastMCP** baseline against the constraints-driven pipeline and selected the final design for lower latency (**~1s** local LoRA vs. **~4.5s** API) and more reliable retrieval in construction workflows.

Real-Time Recommendation System | *Kafka, MLOps, Kubernetes (k3s)*

October 2025

- Integrated with a real-time **Apache Kafka** user-event stream for training and continuous updates; added data quality gates, bad-record quarantine, and feature snapshot drift monitoring with **PSI**.
- Productionized retraining + deployment via **k3s CronJob** with versioned artifacts and atomic model hot-swap for **zero-downtime** rollout and an online **latency gate**; built a hybrid recommender (CBF + UCF) with **RRF/MMR** and evaluated via leave-one-out Hit Rate.

UniNest: Scalable Search Backend | *AWS, Docker, PostgreSQL* | [GitHub Repo](#)

April 2025

- **Hybrid retrieval and ranking:** Combined PostgreSQL **BM25** with semantic vector retrieval and **RRF** fusion, improving Precision@10 by **16%** while maintaining **p95 latency of 329ms**.
- **Deployment and multimodal querying:** Deployed JWT-secured services on AWS EC2 with Docker, enabled image-to-text query flows, and instrumented latency across ETL and search workloads handling **410+ properties/day**.